PRF(A)

Chemical Resistant Fans



systemair

Chemical Resistant Centrifugal Fan Design Casing

The fan casing is constructed from thermoplastic. The fan casing is built to a true volute form and has high efficiency inlet cone to give an even distribution of air over the full width of the runner.

The smaller range of fan sizes, model **PRF(A) 160 - 315** are completely plastic injection moulded and suitable for rotation at any position. The plastic injection moulded backplate or inlet cover can be easily removed for changing of rotation, and for maintenance purposes. The **PRF(A) 160 - 315** casings have circular plain outlet spigots suitable for direct connection to similar sized duct via flexible connection sleeve. Standard casing material is Polypropylene (PP).

Radial Tip Backward Curve Impellers

PRF(A) fan impellers are a Single Inlet Single Width (SISW) type. Impellers are of precision plastic injection moulded design with cast-in metal hub and mechanically welded to highest quality standard with excellent aerodynamic properties.

Impellers are made of PP. Each impeller is statically and dynamically balanced in two planes in accordance with Q2.5 of VDI 2060. The hubs are designed for use with taper-bushes and are made of high-grade cast metal to guarantee reliability at high peripheral speeds.

Fan Support Stand

Fan support stands are precision manufactured from heavy gauge steel. Corrosion prevention is achieved by zinc plating and powder coating as standard fan support stand. The design of the support stand allows rotation of the fan casing to achieve different discharge directions.

Motor

All motors are Totally Enclosed Fan Cooled (TEFC). Standard protection rating is IP55 but can be upgraded to IP56, IP66 or IP66 in stainless steel, upon request. The motors are three phase, 50/60 Hz suitable for 415 volts.

Explosion proof

Explosion proof motors can be supplied upon request.

Standard Colour

All PP - Equivalent to PANTONE Warm Grey 1C

Accessories

- Anti-vibration rubber or spring mounts
- Condensate water drain nipple (standard 25dia on PRF(A) 160 - 315)
- Shower-proof motor cover
- Inlet/Outlet flexible connection sleeves, flanges and duct connection options
- Adjustable-pitch roof brackets and mounting options

Thermal and Chemical Resistance

The temperature of the air and gases must not exceed 80°C for PP material.

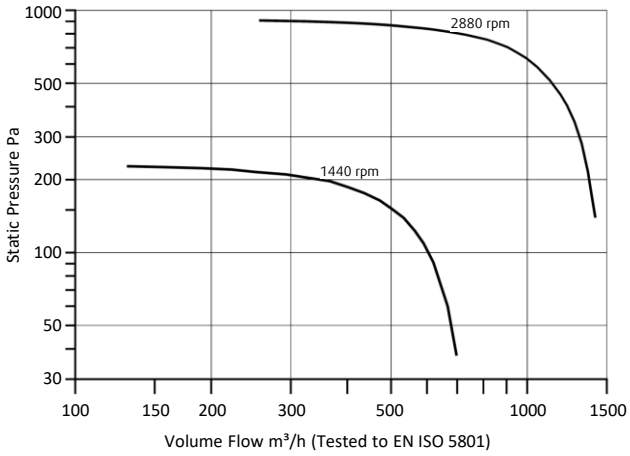
Please contact our sales engineers for chemical resistance information.

Drive Arrangement

PRF(A) fans are supplied in Direct Drive arrangement.

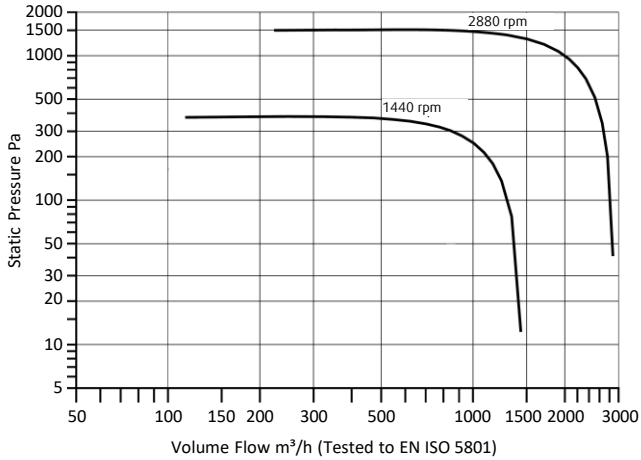


PRF(A) 160



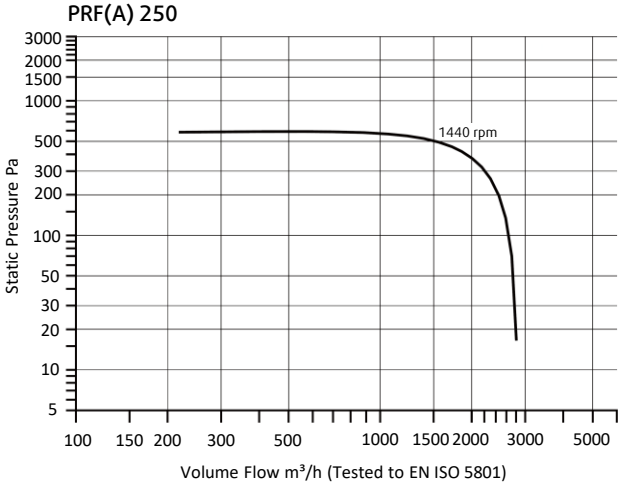
Technical data		1440 rpm D4	2880 rpm D2
Voltage	V	400	400
Frequency	Hz	50	50
Phase	-	3	3
Motor size	kW	0.55	0.75
Full Load Current	A	1.5	1.8
Max. Airflow	m3/h	696	1413
Nominal Speed	RPM	1440	2880
Sound Pressure level Lw (3m)	dB(A)	49	60
Motor Enclosure		TEFC	TEFC
Weight	Kg	34	35

PRF(A) 200



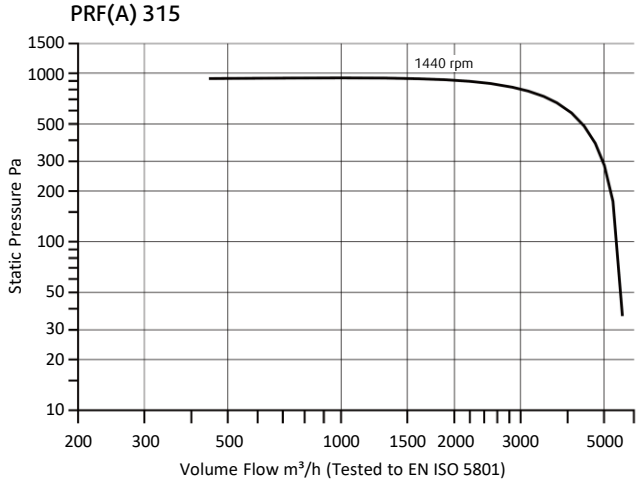
Technical data		1440 rpm D4	2880 rpm D2
Voltage	V	400	400
Frequency	Hz	50	50
Phase	-	3	3
Motor size	kW	0.55	2.2
Full Load Current	A	1.5	4.7
Max. Airflow	m3/h	1438	2870
Nominal Speed	RPM	1440	2880
Sound Pressure level Lw (3m)	dB(A)	53	70
Motor Enclosure		TEFC	TEFC
Weight	kg	37	49

PRF(A) 250



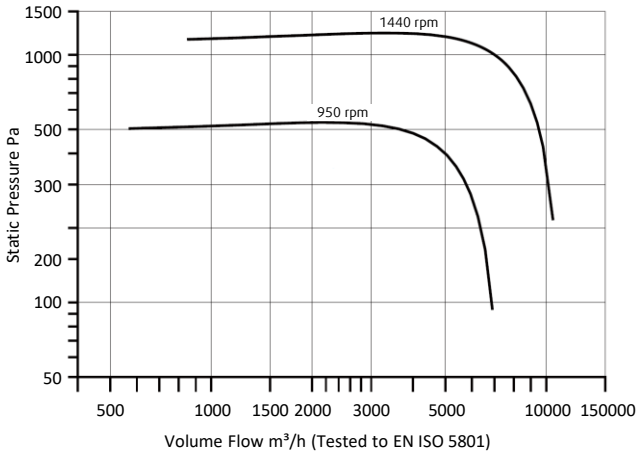
Technical data		1440 rpm D4
Voltage	V	400
Frequency	Hz	50
Phase	-	3
Motor size	kW	1.1
Full Load Current	A	2.7
Max. Airflow	m3/h	2804
Nominal Speed	RPM	1440
Sound Pressure level Lw (3m)	dB(A)	62
Motor Enclosure		TEFC
Weight	kg	40

PRF(A) 315



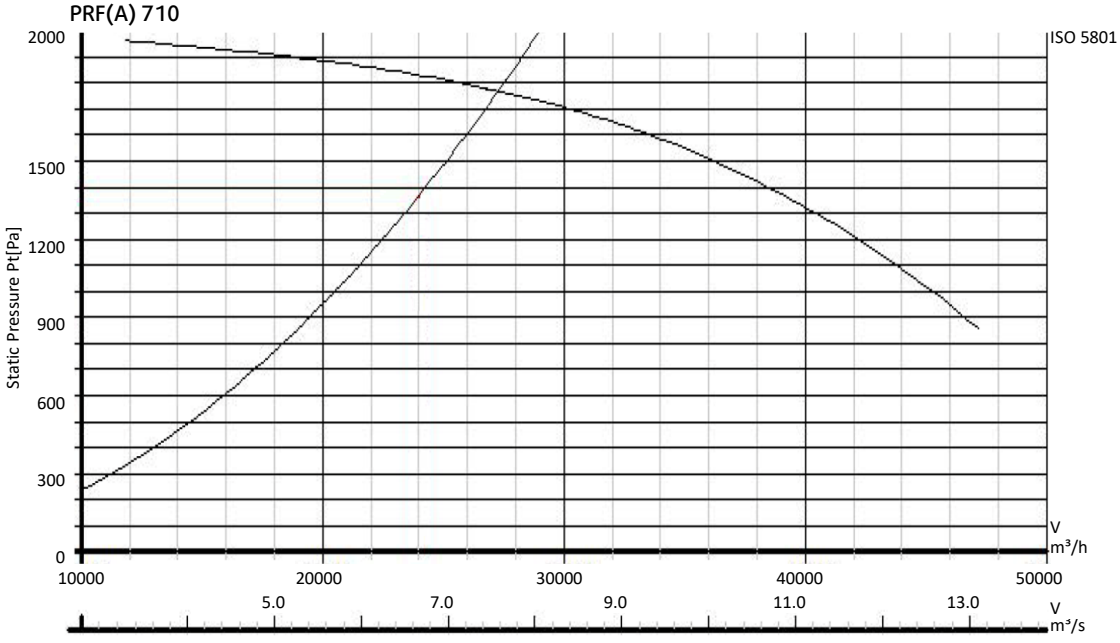
Technical data		1440 rpm D4
Voltage	V	400
Frequency	Hz	50
Phase	-	3
Motor size	kW	3.0
Full Load Current	A	6.5
Max. Airflow	m3/h	5609
Nominal Speed	RPM	1440
Sound Pressure level Lw (3m)	dB(A)	67
Motor Enclosure		TEFC
Weight	Kg	80

PRF(A) 400



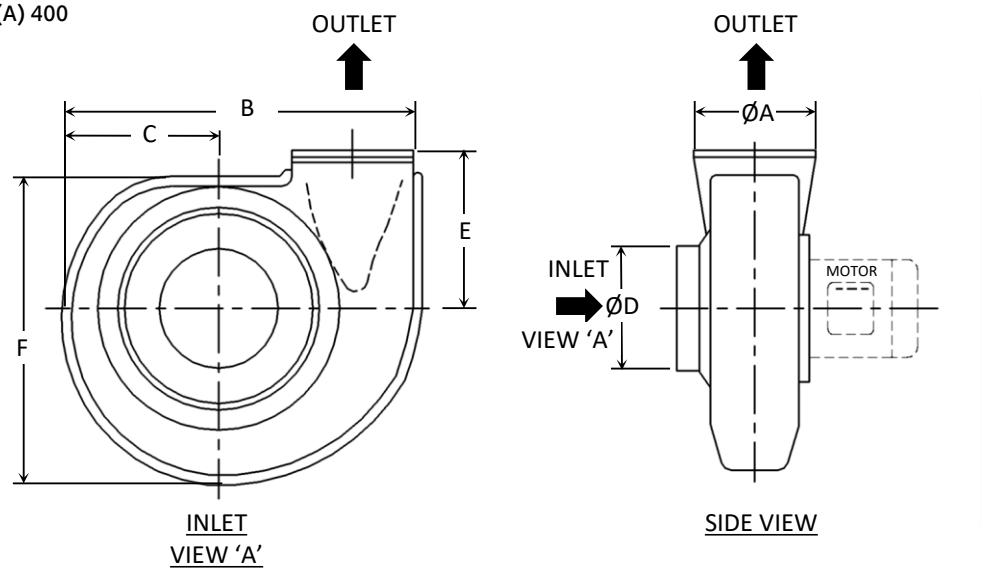
Technical data		1440 rpm D4			950 rpm D6
Voltage	V	400	Voltage	V	400
Frequency	Hz	50	Frequency	Hz	50
Phase	-	3	Phase	-	3
Motor size	kW	5.5	Motor size	kW	2.2
Full Load Current	A	12	Full Load Current	A	5.53
Max. Airflow	m3/h	9215	Max. Airflow	m3/h	6900
Nominal Speed	RPM	1440	Nominal Speed	RPM	950
Sound Pressure level Lw (3m)	dB(A)	73	Sound Pressure level Lw (3m)	dB(A)	67
Motor Enclosure		TEFC	Motor Enclosure		TEFC
Weight	Kg	127	Weight	Kg	106

PRF(A) 710



Technical data		960 rpm D6
Voltage	V	400
Frequency	Hz	50
Phase	-	3
Motor size	kW	30
Full Load Current	A	57.2
Max. Airflow	m3/h	45000
Nominal Speed	RPM	960
Sound Pressure level Lw	dB(A)	88 (1m) 78 (3m)
Motor Enclosure		TEFC
Weight + motor (Approx.)	Kg	572

PRF(A) 160 PRF(A) 200
PRF(A) 250 PRF(A) 315
PRF(A) 400

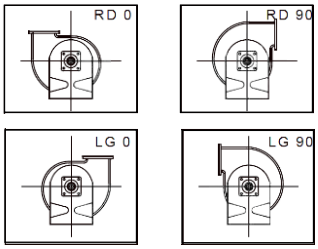


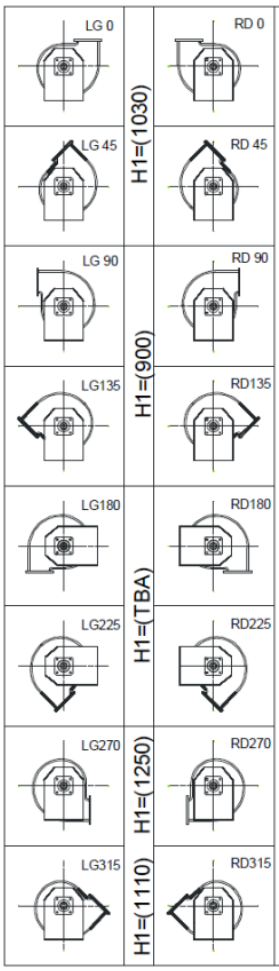
Dimensions

All dimensions in metric (mm)

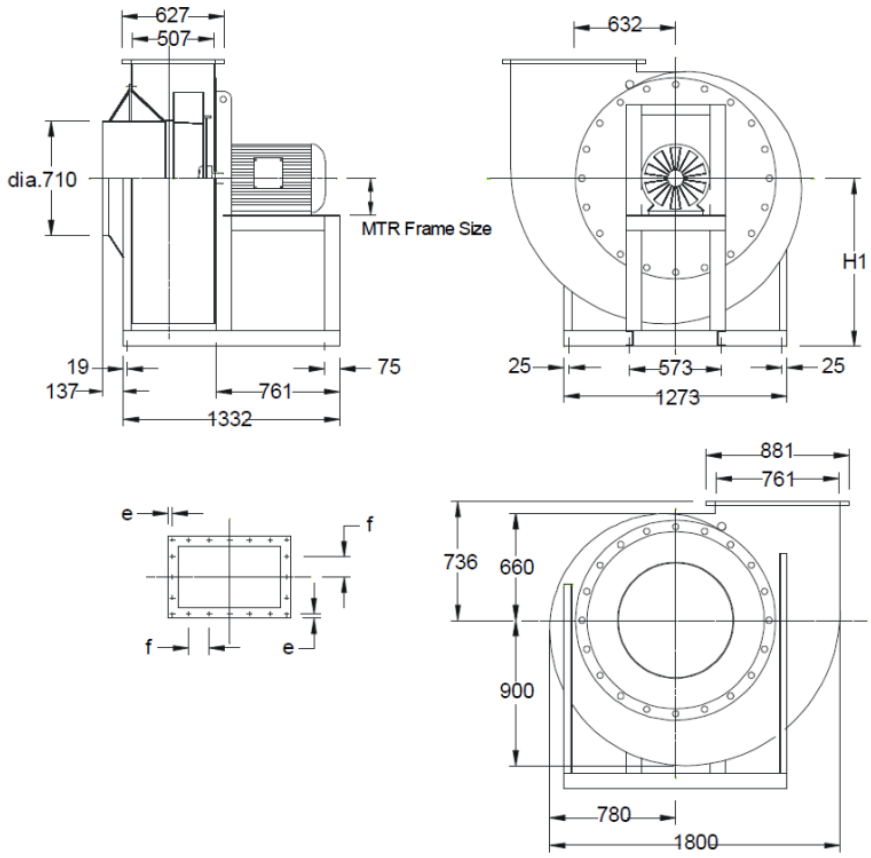
PRF(A)	ØA	B	C	ØD	E	F
PRF(A) 160D4	160	497	223	160	215	429
PRF(A) 160D2	160	497	223	160	215	429
PRF(A) 200D4	200	622	279	200	269	537
PRF(A) 200D2	200	622	279	200	269	537
PRF(A) 250D4	250	703	324	250	320	575
PRF(A) 315D4	315	930	410	315	404	790
PRF(A) 400D4	400	1003	447	400	508	857

Arrangements (View from drive side)





PRF(A) 710



The direction of rotation is as viewed from the drive side.

Barefan weight approx. (280) kg



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